02 DATA TYPES AND VARIABLES • THEORY

03 const constexpr and auto

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

03 — `const`, `constexpr`, and `auto` Type Deduction

- `const`: Promises that the variable cannot be modified after initialization (value may be computed at runtime or compile time).
- `constexpr` **(C++11–C++23)**: Guarantees that the variable's value is a **compile-time constant expression** (usable as an array bound `int arr[MAX];`).
- `auto`: Deduces the variable's type at compile time from its initializer (`auto pi = 3.14;` deduces `double`).